FLIGHT FARE PREDICTION PLATFORM

Clear project summary

64. Flight Fare Prediction Platform

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PROBLEM STATEMENT

Flight prices change often. Travelers may pay more because they do not know when to book. This project predicts future flight prices and tells users when booking may be better.

OBJECTIVES

- Collect old flight price data.
- Predict whether prices may go up or down.
- Suggest a suitable time to book.
- Send an alert when prices fall.
- Show price changes in a simple dashboard.

MODULES

- User Login and Profile
- Flight Price Collection
- Price Prediction
- Price Drop Alerts
- Price Trend Dashboard

DATASET

Kaggle Flight Fare Prediction Dataset

DOMAIN INTEGRATION

- Machine Learning: predict flight prices.
- Web Application: let users track fares.
- Software Engineering: keep each part separate and documented.
- Operating Systems: run price checks automatically and send alerts to many users.

FLIGHT FARE PREDICTION PLATFORM — 7-WEEK DEVELOPMENT PLAN

WEEK 1
START AND COLLECT DATA

BUILD	• Decide what the system must do. • Create project folders. • Download the Kaggle flight price dataset. • Check routes, dates, prices and empty values.
TOOLS	Python, Pandas, Jupyter, VS Code
ARCHITECTURE	Users → Web App → Data Storage
TEST	Open sample records and check the columns.
WORKING OUTPUT	Clean flight price data ready for use.

WEEK 2
COLLECT AND SAVE FLIGHT PRICES

BUILD	• Build a service to collect flight prices. • Use the same format for routes and dates. • Save old prices. • Save user details and preferences.
TOOLS	Python, Pandas, Flask
ARCHITECTURE	User Login → Price Collection → Saved Flight Prices
TEST	Collect → Check → Save → Read data.
WORKING OUTPUT	Working price collection and user profile service.

WEEK 3
PREDICT FLIGHT PRICES

BUILD	• Arrange prices by route and date. • Create useful date and past-price values. • Train a model to predict future prices. • Predict whether the price may rise or fall.
TOOLS	Python, Pandas, scikit-learn, Joblib
ARCHITECTURE	Saved Prices → Prepare Data → Prediction Model → Price Result
TEST	Test predictions using sample routes.
WORKING OUTPUT	Future price direction and prediction result.

FLIGHT FARE PREDICTION PLATFORM — 7-WEEK DEVELOPMENT PLAN

WEEK 4
SUGGEST WHEN TO BOOK

BUILD • Compare today's price with the predicted price. • Create simple rules for booking advice. • Give advice through the web app.

TOOLS Flask, REST API, Python

ARCHITECTURE Price Prediction → Booking Advice → User

TEST Test rising, falling and steady price cases.

WORKING OUTPUT Advice such as Book Now or Wait and Watch.

WEEK 5
SEND PRICE DROP ALERTS

BUILD • Find large price drops. • Run price checks at fixed times. • Send alerts to users. • Avoid sending the same alert again.

TOOLS Python, Scheduler/Cron, Threading or Async

ARCHITECTURE Automatic Price Check → Price Drop Check → Alert → User

TEST Test scheduled checks and several alerts at once.

WORKING OUTPUT Price-drop alerts reach subscribed users.

WEEK 6
BUILD THE DASHBOARD

BUILD • Show old and current prices. • Add route and date filters. • Show predictions and alerts. • Connect the web page to all services.

TOOLS HTML, CSS, JavaScript, Chart Library

ARCHITECTURE Web Page → API → Saved Data → Charts

TEST Compare dashboard values with saved data.

WORKING OUTPUT Simple dashboard for tracking flight prices.

FLIGHT FARE PREDICTION PLATFORM — 7-WEEK DEVELOPMENT PLAN

WEEK 7
FINAL TESTING AND DELIVERY

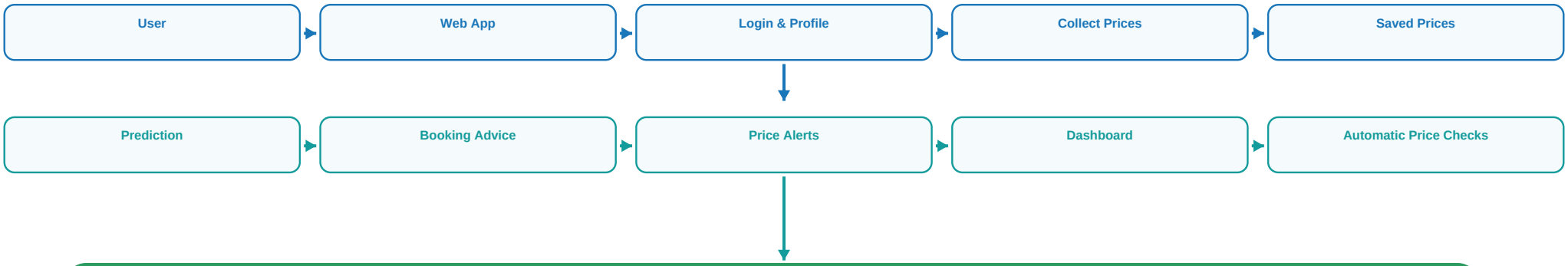
BUILD	• Test different routes and dates. • Check predictions and alerts. • Test the dashboard and services together. • Fix errors and prepare the report.
TOOLS	Pytest, CSV Reports, GitHub
ARCHITECTURE	Test Input → Complete System → Results → Report
TEST	Test route → price data → prediction → advice → alert.
WORKING OUTPUT	Final code, dataset notes, report and demo.

FINAL CHECKLIST

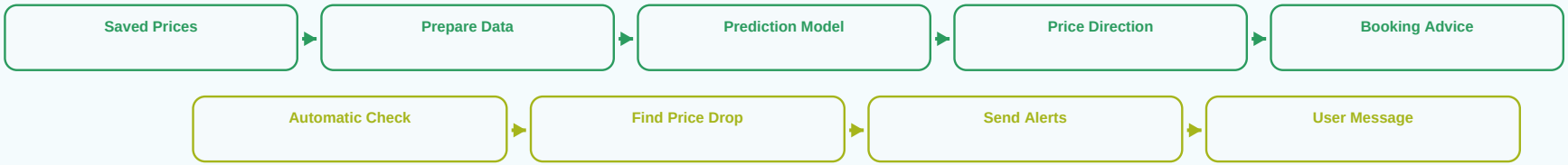
- Collect flight prices by route.
- Predict whether prices may rise or fall.
- Give booking advice.
- Send alerts when prices drop.
- Show price trends on the dashboard.
- Run automatic price checks.
- Send alerts to many users at the same time.
- Keep the project parts separate and documented.

FINAL SYSTEM ARCHITECTURE

Flight Fare Prediction Platform



PRICE PREDICTION AND ALERT FLOW



HOW THE COMPLETE SYSTEM WORKS

User Request → Flight Prices → Prediction → Booking Advice → Alerts → Dashboard